BLOCK 3

One-way (Analysis of Variance) ANOVA

Recap: what do we read from each analysis?

Same dataset, different questions and different numbers to interpret.

ANALYSIS	ASKS…	READ FIRST	P-VALUE
Correlation	Do two numeric measures move together?	Pearson r: sign + strength	$p < .05$ supports a nonzero linear relationship
t-test	Do TWO group means differ?	Means + SDs; t is the test statistic	$p < .05$ supports a difference in means
Regression	How does expected Y change as X changes?	Slope b (coefficient), intercept, SE / CI, R^2	$p < .05$ supports a nonzero slope

The statistic describes the pattern. The p-value evaluates evidence against a no-pattern null.

One p-value idea for all four tests

Imagine there is NO real difference or relationship.

How surprising would a result at least this extreme be, just from random variation?

Example: p = .03

About 3% if the null model were true

p < .05 Conventionally called statistically significant.

p ≥ .05 Insufficient evidence; NOT proof of no effect.

A small p-value does NOT tell us that an effect is large, important, or causal.

Start with variable roles and variable types

Question → variable types → analytical tool

GROUP OR X	OUTCOME OR Y
NUMERIC **baseline_trust**	**NUMERIC** **message_trust**

Two numeric variables → correlation or simple regression

GROUP OR X	OUTCOME OR Y
CATEGORICAL **condition (2 groups)** **education_level (3 groups)**	**NUMERIC** **message_trust**

2 groups + numeric Y → t-test. 3+ groups + numeric Y → ANOVA.

Three analyses you already know

1 CORRELATION



Two numeric variables.
Read r and p.

2 t-TEST



Two groups, numeric Y.
Read means, SD, p.

3 REGREGRESSION



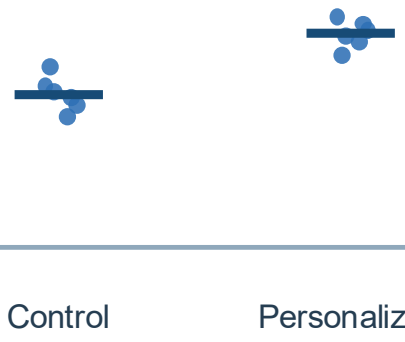
Predict numeric Y.
Read slope, SE / CI, p.

One question is still missing: what if X has THREE or more categories?

The bridge: t-test versus ANOVA

TWO groups

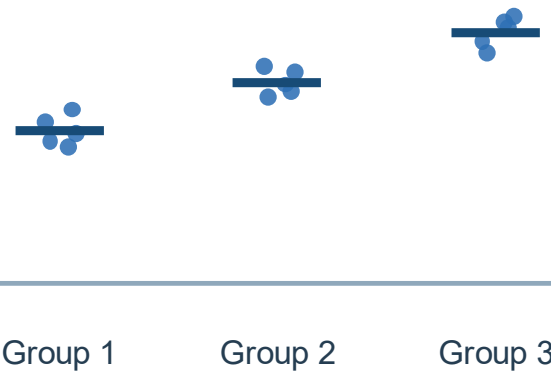
Message trust



t-TEST

THREE+ groups

Message trust



ANOVA

More groups do NOT mean more t-tests. ANOVA starts with ONE overall test.

What is the null hypothesis (H_0)?

STARTING IDEA FOR ANY STATISTICAL TEST

H_0: NO real difference, relationship, or effect

This is the starting claim we test — NOT something we know is true.

t-test / ANOVA: equal averages • Correlation: r = 0 • Regression: slope = 0

Now we look at the p-value:

$p < .05$ **REJECT H_0**

Enough evidence AGAINST the “no pattern” claim.

$p \geq .05$ **FAIL TO REJECT H_0**

Not enough evidence. This does NOT prove “no pattern.”

Do NOT say “accept H_0.” Not finding evidence is NOT proof that nothing exists.

H_0, H_1, H_2: why write hypotheses?

H_1 and H_2 = different predictions we want to check. H_0 = the no-pattern claim for EACH test.

H_1 RESEARCH PREDICTION #1

Personalized messages lead to higher average message trust than Control.

Its H_0: both conditions have the SAME average trust.

H_2 RESEARCH PREDICTION #2

Higher baseline trust is linked to higher message trust.

Its H_0: NO linear relationship between the two trust measures.

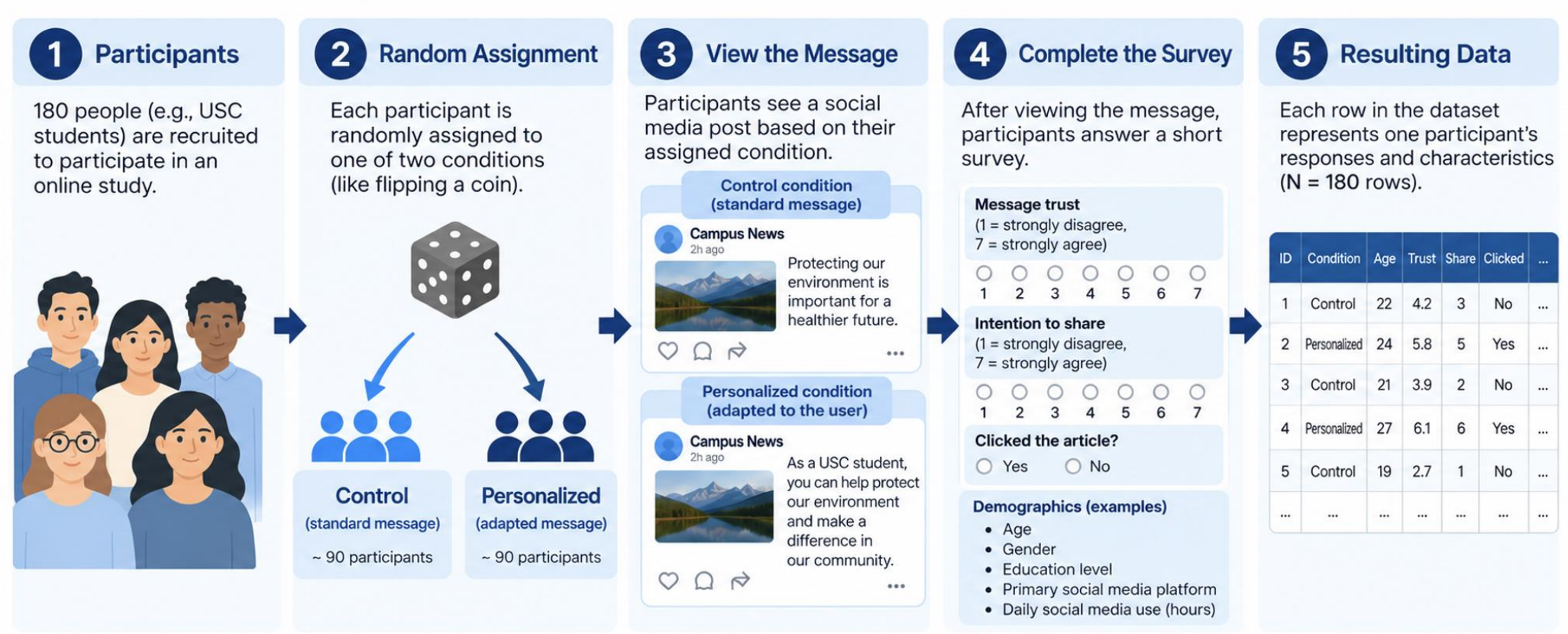
WHY DO WE NEED THEM?

1. **Make a prediction**
2. **Choose the right test**
3. **Evaluate the evidence**

H_1 and H_2 are NOT different test results. Each prediction has a matching H_0.

Same dataset

A randomized experiment with two conditions and a post-message survey (N = 180)



Today we use this SAME dataset to ask three different statistical questions.

The dataset

ROW = one participant / observation

COLUMN = one variable

ID	condition	age	gender	platform	message_trust	clicked
1	Control	24	Woman	TikTok	4.2	No
2	Personalized	31	Man	YouTube	5.8	Yes
3	Control	27	Woman	Instagram	3.9	No

The same dataset can answer different questions depending on which columns we compare.

Let’s look at specific variables

180 **Participants**

one row per participant

2 **Message conditions**

Control / Personalized

3 **Education groups**

HS or some college / Bachelor’s / Graduate

1 **Trust score**

message_trust is numeric

condition: randomized two-group factor | education_level: three observed groups

ANOVA

Research question: **Does average message trust differ by education level?**

X = GROUPING VARIABLE
education_level
3 categories

→

Y = OUTCOME
message_trust
numeric score

ANOVA compares the AVERAGE of Y across the categories of X.

Grouping by education is observational; do not say education CAUSED trust.

What does ANOVA actually compare?

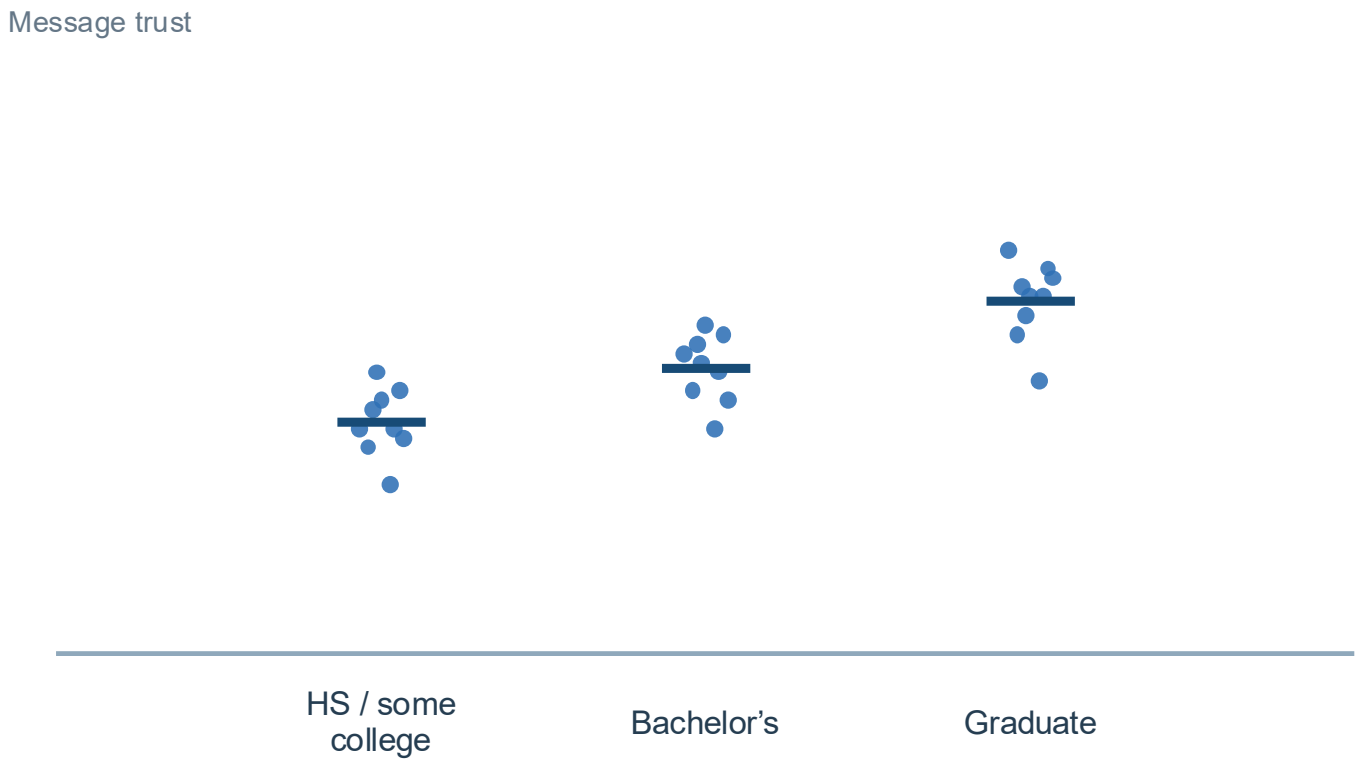


Illustration only — Colab generates the actual class plot.

Each dot

= one participant

Dark horizontal line

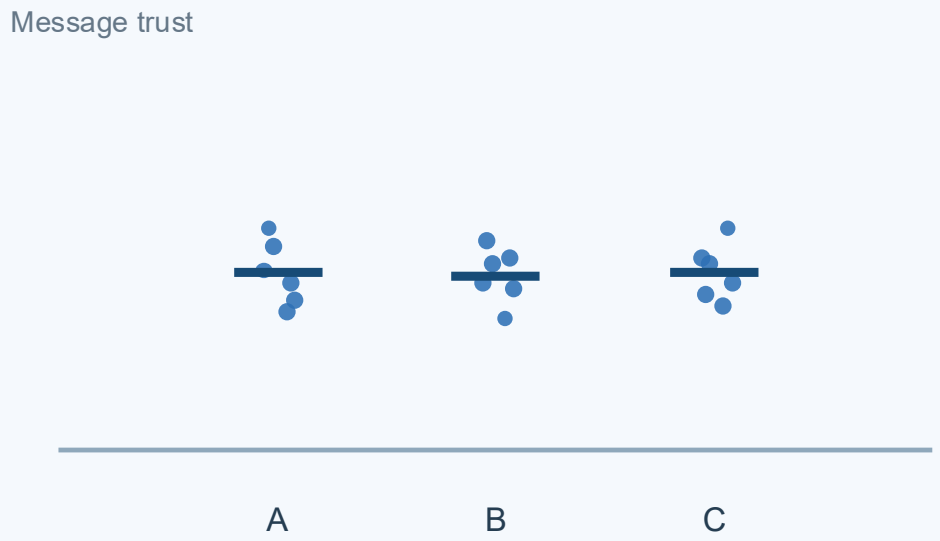
= that group's mean

Question:

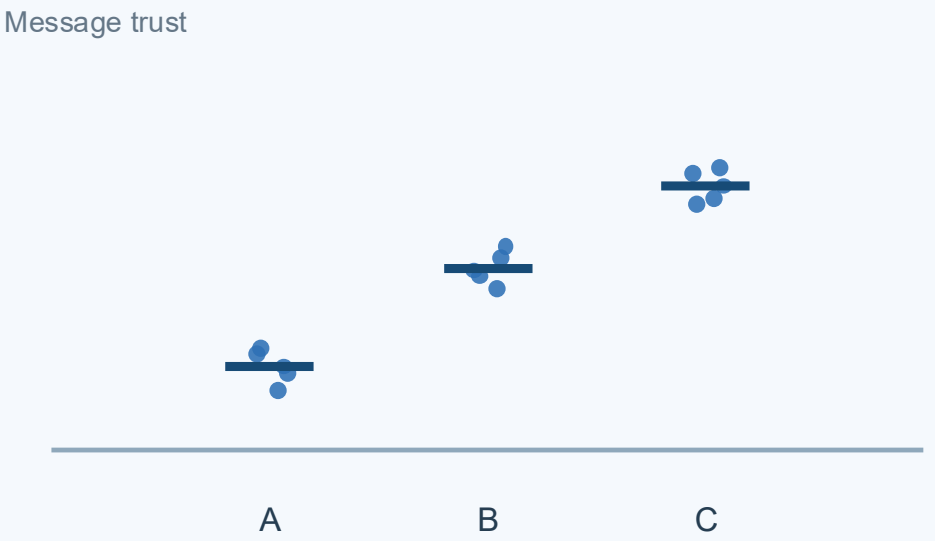
Are these average differences bigger than ordinary person-to-person variation?

The idea: differences BETWEEN groups versus WITHIN groups

Small between-group difference



Large between-group difference



ANOVA weighs separation of the means against the spread within each group.

F is a ratio

$$F = \frac{\text{Variation BETWEEN group means}}{\text{Variation WITHIN groups}}$$

More separation relative to spread → usually a bigger F

F alone does not tell you WHICH specific groups differ. Read p, then follow-ups.

The ANOVA question: are ALL group means equal?

NULL HYPOTHESIS (H_0)
All three POPULATION group means are equal

ALTERNATIVE (H_1)
AT LEAST ONE population group mean differs

$p < .05 \rightarrow$ evidence against H_0

$p \geq .05 \rightarrow$ insufficient evidence

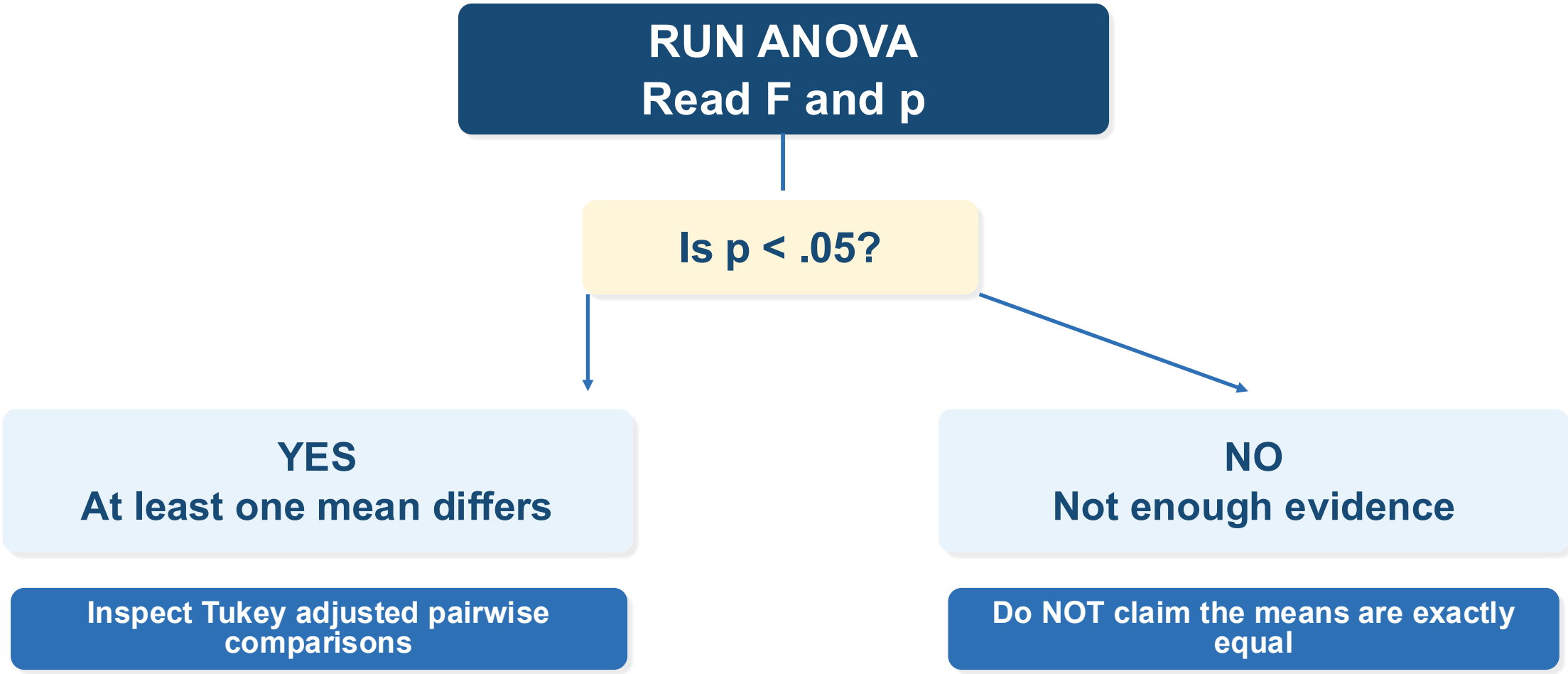
ANOVA is an OVERALL (omnibus) test; it does not name the differing pairs.

What should we read from an ANOVA?

1	**MEAN**	Which group's average is higher or lower?
2	**SD + n**	How spread out are the scores, and how many people?
3	**F + p**	Is there overall evidence that not all means are equal?
4	**TUKEY**	If needed, which specific pairs differ?
5	**η^2 (OPTIONAL)**	How much sample variation is associated with group?

First describe the groups. Then interpret the statistical evidence.

After ANOVA: what happens next?



Open the Colab notebook: now YOU run ANOVA

1	**Open**	colab.research.google.com
2	**Choose**	File → Upload notebook
3	**Select**	JOUR701_ANOVA_Colab_Lab.ipynb
4	**Run**	Click ▶ beside each code cell in order

Use the NOTEBOOK to run and interpret; the slides are your visual guide.

Work at your own pace. Ask what the output MEANS before worrying about code.

Colab steps 1–2: load and clean the dataset


```
import pandas as pd

raw = pd.read_csv(url)  # same link as last class
df = raw.drop_duplicates("participant_id")

df["message_trust"] = pd.to_numeric(
    df["message_trust"], errors="coerce")

anova_df = df[["education_level",
               "message_trust"]].dropna()
```

What Python does

- Reads our original CSV
- Keeps one row per participant
- Uses the 2 columns we need
- Leaves out missing values

Colab step 3: read the means and SDs FIRST


```
summary = (anova_df.groupby("education_level")
        ["message_trust"]
        .agg(n="count", mean="mean", SD="std"))
display(summary.round(2))
```

READ THE OUTPUT

- n = people in each group
- Mean = average trust
- SD = how spread out scores are

DISCUSS: Which group has the highest SAMPLE mean? Is the gap large compared with the SD?

A sample difference is not automatically a statistically significant difference.

Colab step 4: visualize the three groups


```
# In the notebook:
# one dot = one person
# dark line = group mean

plt.scatter(x, y, alpha=0.45)
ax.hlines(y.mean(),
          i - 0.22, i + 0.22)
plt.show()
```

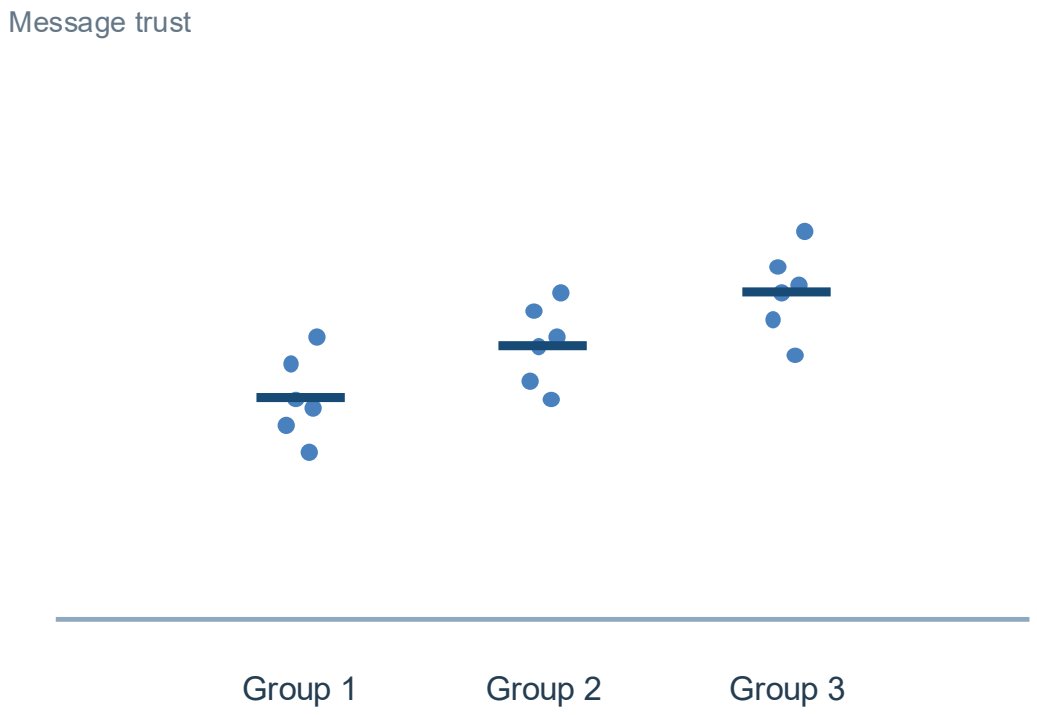


Illustration — use the notebook for real values.

LOOK FOR: group centers • within-group spread • overlap • extreme values

Colab step 5: run ONE ANOVA


```
import statsmodels.api as sm
import statsmodels.formula.api as smf

model = smf.ols(
  "message_trust ~ C(education_level)",
  data=anova_df
).fit()

anova_table = sm.stats.anova_lm(
    model, typ=2)
display(anova_table)
```

Read it like English

- Left of ~: numeric outcome
- Right of ~: category predictor
- C( ) means “treat as groups”
- Read F and PR(>F)

You do NOT need every number in the ANOVA table

Find the row labeled C(education_level):

SOURCE	F	PR(>F)	WHAT IT MEANS
C(education_level)	**YOUR F**	**YOUR p**	p < .05? → overall difference
Residual	—	—	Differences left within the groups

F = between-group variation relative to within-group variation

Use the p-value to decide whether there is evidence of ANY group mean difference.

Colab step 6: Tukey shows which pairs differ


```
from statsmodels.stats.multicomp import (
    pairwise_tukeyhsd)

tukey = pairwise_tukeyhsd(
  endog=anova_df["message_trust"],
  groups=anova_df["education_level"],
  alpha=0.05
)
print(tukey.summary())
```

What to look for

- group1, group2 = the pair
- meandiff = difference in averages
- p-adj = adjusted p-value
- reject = True means pair differs

ANOVA asks “any difference?”
Tukey asks “which pair?”

Now write ONE clear ANOVA result

APPLY

Fill in from your own Colab output:

We compared average message trust across three education groups.

The group means (and SDs) were [write all three].

The overall ANOVA was [significant / not significant], F(df1, df2) = [ ], p = [ ].

[If relevant: Tukey indicated which pairs differed.]

Keep description separate from claims about statistical significance or causation.

Partner challenge: choose the analysis

APPLY

baseline_trust and message_trust: are they related?	**Correlation**
Control vs Personalized: do mean trust scores differ?	**t-test**
How much does expected trust change per 1-unit baseline trust?	**Regression**
Does mean trust differ by three education levels?	**ANOVA**

Start with the QUESTION and variable TYPES, not the test name.

Day 2 checkpoint: explain ANOVA in your own words

APPLY

- Why ANOVA instead of a t-test for education_level?
- Which variable is the categorical X? Which is numeric Y?
- What do a group mean, SD, F, and p tell you?
- What can you learn from Tukey after an overall difference?
- Can you conclude education CAUSES message trust?

If you can answer these in plain language, you understand the core logic.